SOLUTION REQUIREMENTS

Comic Craft – AI Comic Story Creator using Gemini Models

Date	15 March 2026
Team ID	SWTID-2026-8237
Project Name	Comic Craft – AI Comic Story Creator using Gemini Models

Maximum Marks: 4 Marks

Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do). Non-Functional Requirements (NFR) specify criteria that judge the operation of the system (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	The current Comic Craft version does not implement user authentication or account login. The application can be accessed locally through the web interface.	Low
2	**Authorization levels**	No role-based authorization is implemented in the current version. The project documentation states that the current version focuses on single comic generation without user account management.	Low
3	**External interfaces**	The system integrates with Google Gemini models for story generation and Hugging Face Diffusers/Stable Diffusion for comic illustrations. FastAPI exposes browser and JSON API routes, and	High

		Jinja2 renders the frontend.	
4	**Transactions processing**	The core transaction is comic generation: receive prompt and preferences, generate a 5-panel outline, generate narration/dialogue, generate an image for each panel, build the layout, and export the comic as a PDF. No payment transaction is described.	High
5	**Reporting**	The system provides a generated comic preview and an exportable PDF containing panel images, titles, scene descriptions, captions and narration. A separate analytical/reporting dashboard is not described.	Medium
6	**Business rules, etc.**	User inputs include story prompt, character name, setting, story tone and art style. The standard workflow generates a structured 5-panel comic. Changing tone or art style regenerates the outline, story and images with corresponding model instructions.	High
7	**Compliance to laws or regulations**	No specific legal, industry or privacy standard is specified in the supplied project documentation. API keys and Hugging Face	Medium

		tokens are to be stored using environment variables rather than hard-coded.	
8	**Other**	The application provides /generate for form-based generation, /generate-comic/json for JSON API generation, /test-image for image-generation testing, and /export-success for export confirmation.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	The application is intended to provide an efficient automated comic-generation workflow using Gemini Flash for structured outline generation and Stable Diffusion for image generation. Exact response-time targets are not specified in the project documentation.	Acceptance: complete generation workflow should execute successfully; numerical response-time target: To be defined.
2	**Scalability**	The documented deployment is a local Fast API application. The architecture is modular, with separate frontend, backend and AI integration components, but a production concurrency target is not specified.	Acceptance: application runs in the documented local environment; production concurrent-user target: To be defined.
3	**Security & Data Privacy**	Sensitive Gemini and Hugging Face credentials are configured through environment variables/.env rather than being embedded directly in application code. The supplied documentation does not specify encryption standards or a formal vulnerability-management process.	Acceptance: API credentials are supplied through environment configuration; encryption/compliance target: To be defined.
4	**Reliability & Availability**	Fast API/Unicorn provides the documented local server workflow. Error	Acceptance: expected routes operate and failures are handled

		handling uses HTTP Exception and try-except blocks for unexpected failures such as API or JSON decoding errors.	gracefully; uptime target: To be defined.
5	**Usability & Accessibility**	The interface is designed to be simple and user-friendly, with labeled fields for story prompt, character name, setting, tone and art style. The comic preview presents panels sequentially and the export flow confirms completion.	Acceptance: users can enter inputs, preview generated panels and export the comic through the documented UI. Formal WCAG target: To be defined.
6	**Other – Maintainability & Portability**	The project uses a modular structure with separate modules for Gemini Flash outline generation, Gemini Pro story generation, image generation, layout building, PDF export, Fast API routes and Jinja2 templates. A virtual environment and requirements.txt are used for dependency management.	Acceptance: components can be installed and launched using the documented environment setup and Unicorn command.

Project Requirement Summary

Requirement Area	Comic Craft Specification
System Purpose	Generate personalized comic stories and illustrations from user prompts through a web interface.
Input	Story prompt, main character name, setting, story tone, and art style.
AI Story Generation	Gemini Flash generates a structured 5-panel outline; Gemini Pro generates detailed narration and dialogue.
AI Image Generation	Stable Diffusion generates comic-style illustrations from panel image prompts.
Backend	Fast API manages routing, user input, AI workflow coordination, layout creation and PDF export.
Frontend	HTML, CSS and Jinja2 templates provide input, comic preview and export-success pages.
Output	Panel-by-panel comic preview and downloadable multi-page PDF.
Deployment	Local deployment using Python virtual environment and Unicorn; API documentation is available through Fast API /docs.

Source basis: Comic Craft project document and the supplied Solution Requirements template. Items marked “To be defined” are not specified in the supplied project documentation.